DevOps Infrastructure – Unit 1 Notes (Part 5)

Docker Architecture, Docker Daemon & Docker CLI

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1. Docker Architecture

Definition:

The **Docker Architecture** defines how different components of Docker interact to:

- Build images
- Manage containers
- Execute commands

Main Components of Docker Architecture

Docker follows a **client-server architecture** consisting of:

1. **Docker Client**
2. **Docker Daemon (Server)**
3. **Docker Registry**
4. **Docker Objects (Images, Containers, etc.)**

Architecture Diagram (Conceptual)

User → Docker CLI → Docker Daemon → Containers

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Docker Registry

Working Flow

1. User runs a command using CLI
2. CLI sends request to Docker Daemon
3. Daemon processes request
4. If needed, pulls image from registry
5. Container is created and executed

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2. Docker Daemon

Definition:

The **Docker Daemon** (dockerd) is the core background service that:

- Manages Docker objects
- Handles container lifecycle
- Communicates with the OS

Key Responsibilities

Container Management

- Create, start, stop, delete containers

Image Management

- Build and store images

Networking

- Configure container communication

Storage Management

- Handle volumes and file systems

Important Points

- Runs in the background
- Listens for API requests
- Can communicate with other daemons

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3. Docker CLI (Command Line Interface)

Definition:

The **Docker CLI** is the tool used by users to interact with Docker.

Role of CLI

- Sends commands to Docker Daemon
- Provides user-friendly interface
- Executes operations like:
    - Build
    - Run
    - Stop containers

Common Commands

Check Docker Version

docker --version

List Running Containers

docker ps

List All Containers

docker ps -a

Stop Container

docker stop <container_id>

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4. Interaction Between CLI and Daemon

Communication Method

- CLI uses **REST API**
- Communicates via:
    - UNIX socket (Linux)
    - TCP (remote systems)

Example Flow

Command:

docker run nginx

Internally:

1. CLI sends request to daemon
2. Daemon checks local image
3. If not found → pulls from registry
4. Creates and starts container

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5. Docker Registry in Architecture

- Stores images (e.g., Docker Hub)
- Daemon pulls/pushes images from/to registry

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6. Advantages of Docker Architecture

- Modular design
- Scalable
- Efficient communication
- Easy automation

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7. Summary

- Docker uses client-server architecture
- Docker CLI = user interface
- Docker Daemon = core engine
- Registry = image storage system
- CLI communicates with daemon via API